Explicit sequences

1 Identify the following sequences as finite or infinite:

a $1, 2, 3, 4, 5, 6$

b $-1, -2, -3, -4, -5, -6, \ldots$

2 What is the third term in the sequence $4, -5, 6, -7, 8$?

3 Find the next number in the sequence $16, 36, 64, 100, 144, \ldots$

4 Find the first five terms of the following sequences. Round the terms to four decimal places where necessary.

a $u_n = \dfrac{n+4}{n+3}$

b $u_n = \dfrac{5n-1}{n^2+5}$

c $u_n = \left(\dfrac{1}{2}\right)^n (n-3)$

d $u_n = 3n - 3$

e $u_n = -2^n$

f $u_n = (-2)^n (3n)$

g $u_n = \dfrac{1}{\sqrt{n}}$

h $u_n = \left(1 + \dfrac{1}{n}\right)^n$

5 For the sequence $3, 0.3, 0.03, 0.003, \ldots$, write a formula for the nth term of the sequence, u_n, in terms of n.

Applications

6 Christa deposits $\$1$ into a new bank account while Jenny deposits $\$38$ into a new bank account. The next day, Christa adds $\$2$ to the account, and doubles how much she adds each day thereafter. Jenny adds $\$5$ each day.

a Fill in the amount Christa had in the account each day for the first four days.

b Fill in the amount Jenny had in the account each day for the first four days.

c On which day will they have the same amount of money?

7 For a photo, the staff at a company have been arranged such that there are 9 people in the front row and each row has 2 more people than in the row in front of it.

a Write a formula for u_n in terms of n.

b How many people are in the 10th row?

8 A basketball is dropped onto the ground from a height of 15 metres. On each bounce, the ball reaches a maximum height of 55% of its previous maximum height.

a Write a formula for u_n, for the height reached on the nth bounce in terms of n.

b How high does the basketball reach after the 5th bounce? Round your answer to two decimal places.